Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

The Rocket's Journey — ANSWERS

How a model rocket flies

Every model rocket flight follows the same pattern. The rocket starts out sitting still on the launch pad, waiting for the trigger. When the engine ignites, the rocket lifts off and is pushed hard upwards. This thrust only lasts a second or two, until the fuel runs out.

The rocket does not stop there. It keeps flying upwards, coasting on the speed it has built up, but with nothing pushing it any more it slows down, until for an instant it hangs at its highest point: the peak.

Then it falls. It speeds up at first, but the faster it falls, the harder the air pushes back, until it is falling at a steady speed, its terminal velocity. Next, a small charge in the engine pops out the parachute. The parachute catches the air and slows the rocket right down to a new, much slower terminal velocity, and it drifts gently back to the ground and lands, ready to fly again.

Every part of that story is about forces. Today you turn it into a velocity–time graph, and read off where the forces are balanced and where they are not.

Balanced and unbalanced forces

The rule

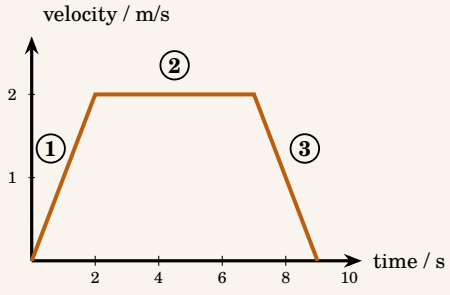
Balanced forces (resultant force = 0): the velocity stays the **same**. A still object stays still; a moving object keeps the same speed in the same direction.

Unbalanced forces (resultant force not zero): the velocity **changes**, in the direction of the resultant force. With the motion, the object speeds up; against the motion, it slows down.

On a velocity–time graph: a **flat** line means balanced forces; a **sloping** line means unbalanced forces. The steeper the slope, the bigger the resultant force.

Worked example — copy this shape

A lift carries you from the ground floor up to the top floor. Two forces act on the lift: the cable pulling it up, and its weight pulling it down.



①	cable force > weight	**unbalanced**, upwards: speeds up
②	cable force = weight	**balanced**: steady speed, 2 m/s
③	cable force < weight	**unbalanced**, downwards: slows down

What just happened. Each straight part of the line got a number, and each number got its forces. The flat part is the only balanced part. *You can feel this:* in ① the floor pushes up on you harder than usual and you feel heavier; in ③ you feel lighter; in ② you feel normal, even though you are moving.

Check it. Every flat part of a graph pairs with "balanced"; every sloping part pairs with "unbalanced", and the resultant points the way the velocity is changing.

The rocket's journey

Before you start

Up is positive. While the rocket rises its velocity is positive and the line is **above** the time axis. While it falls its velocity is negative and the line is **below** the axis.

Air resistance always acts **against** the direction of motion, and it grows as the rocket moves faster. When the rocket is not moving, there is none.

Two simplifications: the rocket flies straight up and comes straight back down • the rocket's mass stays the same.

Read the whole story first. Then sketch it.

① **Waiting for the trigger (0–1 s).** The rocket sits still on the launch pad.

② **Lift-off (1–3 s).** The engine fires. The rocket speeds up from 0 to 40 m/s, upwards.

③ **Coasting (3–6 s).** The fuel runs out and the thrust stops. The rocket is still moving upwards, but slows down from 40 m/s to 0.

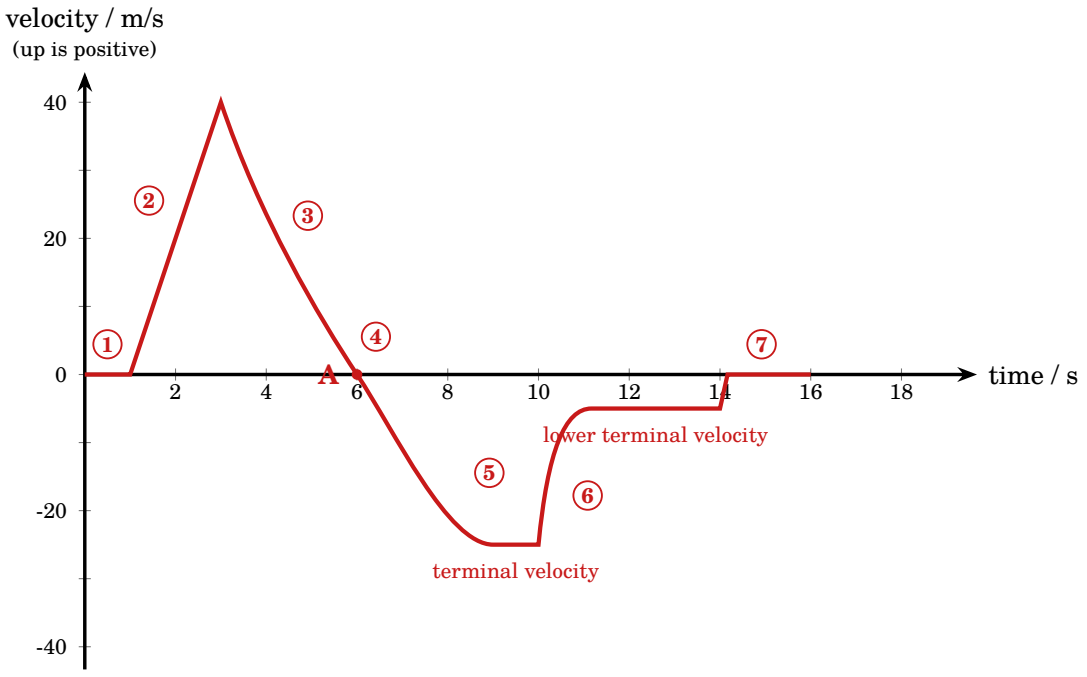
④ **The peak (6 s).** For an instant the rocket is not moving: it is at its highest point. Label this point **A**.

⑤ **Falling (6–10 s).** The rocket falls. It speeds up, but less and less quickly, until from about 9 s it falls at a steady 25 m/s: its **terminal velocity**.

⑥ **Parachute (10–14 s).** At 10 s the parachute ejects. The rocket slows down, quickly at first and then more gently, until it falls at a new, much lower terminal velocity of 5 m/s.

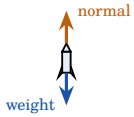
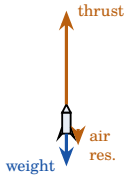
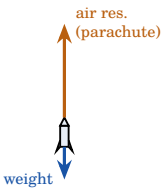
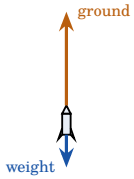
⑦ **Landing (14 s).** The rocket lands, and its velocity returns to 0.

Task 1 — On the **graph paper sheet**, sketch the velocity–time graph of the whole journey. The time axis matches the story, second by second. Number each section ① to ⑦ to match the story, and label point **A**.



Check it. Your line starts and ends at zero, sits above the axis while the rocket rises and below it while it falls, and crosses the axis once in the air, at **A**. Count your flat sections: each one is a part of the story where the velocity does not change.

Task 2 — For each section, draw the forces on the rocket (arrow length shows size), name them, and say whether they are balanced.

#	Force diagram	Forces acting	Balanced or unbalanced?	Velocity: speeding up, slowing down, or steady?
①		weight (down); normal contact force from the pad (up)	balanced	steady (still, 0 m/s)
②		thrust (up), bigger than weight and air resistance (both down)	unbalanced: resultant up	speeding up, upwards
③		weight and air resistance, both down	unbalanced: resultant down	slowing down (still rising)
④		weight only: not moving, so no air resistance	unbalanced: resultant down	zero for an instant, changing from up to down
⑤		weight (down); air resistance (up), growing as it speeds up	unbalanced (down) at first; balanced once air resistance = weight	speeding up less and less, then steady at 25 m/s: terminal velocity
⑥		weight (down); air resistance (up), much bigger once the parachute opens, then falling back	unbalanced (up) at first; balanced at the new terminal velocity	slowing down (still falling), then steady at 5 m/s
⑦		push of the ground (up), much bigger than weight; then equal once still	unbalanced (up) while stopping; balanced once still	slowing to a stop, then steady at 0

What does your graph tell you?

(a) At point **A** the rocket's velocity is zero for an instant. Are the forces on it balanced at that moment? Use your force diagrams to explain.

No. At A the rocket is not moving, so there is no air resistance and the thrust has stopped: weight is the only force. The resultant force is downwards, not zero. The velocity is zero only for an instant: it is changing, from up to down.

[2]

(b) Compare the slope of your line in section ③ with its slope near the end of section ⑤, just before it goes flat. Explain the difference using the forces.

Section 3 is steeper. Rising, air resistance acts downwards, the same way as weight, so the two add to a large resultant and the velocity changes quickly. Falling, air resistance acts upwards against weight, so the resultant is smaller; as air resistance grows the slope gets shallower, until the line is flat at terminal velocity.

[3]

(c) Sections ⑤ and ⑥ both end in a flat line.

(i) What does a flat line tell you about the forces on the rocket?

(ii) Explain why the rocket's steady speed is so much lower in section ⑥.

(i) The forces are balanced: air resistance equals weight, so the velocity does not change. This steady speed is the terminal velocity. (ii) The open parachute has a much bigger area, so the air resistance is much bigger at any speed. It balances the weight at a much lower speed.

[3]

(d) **Stretch.** The rocket's mass is 0.1 kg. In section ② it speeds up by 20 m/s every second. Use $F = ma$ to calculate the resultant force on it during lift-off.

$F = ma = 0.1 \times 20 = 2$ N, upwards. (This is the resultant of the thrust, the weight and the air resistance.)

[2]